



| Abstract POs defined by National Board of Accreditation |                                            |      |                                |
|---------------------------------------------------------|--------------------------------------------|------|--------------------------------|
| PO#                                                     | Broad PO                                   | PO#  | Broad PO                       |
| PO1                                                     | Engineering Knowledge                      | PO7  | Environment and Sustainability |
| PO2                                                     | Problem Analysis                           | PO8  | Ethics                         |
| PO3                                                     | Design/Development of solutions            | PO9  | Individual and team work       |
| PO4                                                     | Conduct investigations of complex problems | PO10 | Communication                  |
| PO5                                                     | Modern tool usage                          | PO11 | Project Management and Finance |
| PO6                                                     | The Engineer and Society                   | PO12 | Lifelong learning              |

### Assessment Pattern

| Bloom's Category | Continuous Assessment Tests |                    | End Semester Examination Marks |
|------------------|-----------------------------|--------------------|--------------------------------|
|                  | Test1 (percentage)          | Test2 (percentage) |                                |
| Remember         | 40                          | 40                 | 40                             |
| Understand       | 40                          | 40                 | 40                             |
| Apply            | 20                          | 20                 | 20                             |
| Analyze          |                             |                    |                                |
| Evaluate         |                             |                    |                                |
| Create           |                             |                    |                                |

### Mark Distribution

| Total Marks | CIE Marks | ESE Marks | ESE Duration |
|-------------|-----------|-----------|--------------|
| 150         | 50        | 100       | 3            |

### Continuous Internal Evaluation Pattern:

|                                                              |          |
|--------------------------------------------------------------|----------|
| Attendance                                                   | 10 marks |
| Continuous Assessment Tests(Average of Internal Tests 1 & 2) | 25 marks |
| Continuous Assessment Assignment                             | 15 marks |

**Internal Examination Pattern:****ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Each of the two internal examinations has to be conducted out of 50 marks. First Internal Examination shall be preferably conducted after completing the first half of the syllabus and the Second Internal Examination shall be preferably conducted after completing the remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly covered module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly covered module), each with 7 marks. Out of the 7 questions in Part B, a student should answer any 5.

**End Semester Examination Pattern:**

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 questions from each module of which a student should answer any one. Each question can have a maximum of 2 subdivisions and carries 14 marks.

**SYLLABUS****Module 1(Introduction to basic concepts and Recent developments )**

Introduction to basic concepts and Recent developments , Collaborative recommendation -User-based nearest neighbor recommendations, Item-based nearest neighbour recommendation, Collaborative recommendation ratings , Model-based and preprocessing-based approaches , Recent practical approaches and systems Content-based recommendation - Content representation and content similarity Similarity-based retrieval and Other text classification methods

**Module 2 (Knowledge-based recommendation)**

Knowledge-based recommendation - Knowledge representation and reasoning , Constraints, Cases and similarities, Interacting with constraint-based recommenders - Defaults Dealing with unsatisfiable requirements and empty result set, Proposing repairs for unsatisfiable requirements, Ranking the items/utility-based recommendation, Interacting with case-based recommenders, Critiquing -Compound critiquing, Dynamic critiquing

**Module 3 (Hybrid recommendation approaches )**

Hybrid recommendation approaches - Opportunities for hybridization Recommendation paradigms, Hybridization designs, Monolithic hybridization design - Feature combination hybrids, Feature augmentation hybrids, Parallelized hybridization design - Mixed hybrids, Switching hybrids, Weighted hybrids, Pipelined hybridization design Cascade hybrids, Meta-level hybrids Limitations of hybridization strategies

## Module 4 (Evaluating Recommender Systems)

Introduction - Evaluation Paradigms , User Studies , Online Evaluation Offline Evaluation with Historical Data Sets, General Goals of Evaluation Design - Accuracy, Coverage , Confidence and Trust , Novelty , General Goals of Evaluation Design - Serendipity ,Diversity , Robustness and Stability Scalability, Design Issues in Offline Recommender Evaluation - Case Study of the Netflix Prize Data Set , Segmenting the Ratings for Training and Testing - Hold-Out , Cross-Validation , Comparison with Classification, Accuracy Metrics in Offline Evaluation - Measuring the Accuracy of Ratings Prediction , RMSE versus MAE, Impact of the Long Tail, Evaluating Ranking via Correlation , Evaluating Ranking via Utility Evaluating Ranking via Receiver Operating Characteristic, Limitations of Evaluation Measures - Avoiding Evaluation Gaming

## Module 5 (Attack-Resistant Recommender Systems)

Introduction Understanding the Trade-Offs in Attack Models - Quantifying Attack Impact Types of Attacks - Random Attack . Average Attack Bandwagon, Reverse Bandwagon Attack, Probe Attack Segment Attack, Effect of Base Recommendation Algorithm, Detecting Attacks on Recommender Systems - Individual Attack Profile Detection , Group Attack Profile Detection - Preprocessing Methods Online Methods Strategies for Robust Recommender Design - Preventing Automated Attacks with CAPTCHAs Using Social Trust. Designing Robust Recommendation Algorithms - Incorporating Clustering in Neighborhood Methods Fake Profile Detection during Recommendation Time Association - Based Algorithms

### Text Books

1. Jannach D., Zanker M. and FelFering A., Recommender Systems: An Introduction, Cambridge University Press(2011)
2. C.C. Aggarwal, Recommender Systems: The Textbook, Springer, 2016.

### Reference Books

1. F. Ricci, L Rokach, B. Shapira and P.B. Kantor, Recommender systems handbook, Springer 2010
2. Manouselis N., Drachsler H., Verbert K., Duval E., Recommender Systems For Learning, Springer (2013), 1st ed.

### Course Level Assessment Questions

**Course Outcome 1 (CO1):** Discuss the cases in which content-based recommendations will not perform as well as collaborative filtering.

**Course Outcome 2 (CO2):** Analyze, in detail, different techniques available to support users in the interaction with constraint-based recommender applications.

**Course Outcome 3(CO3):** Explain about the feature combination and feature augmentation hybrid mechanisms.

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**Course Outcome 4 (CO4):** With appropriate case study, explain the design issues in offline recommender evaluation .

**Course Outcome 5 (CO5):** Illustrate different methods used to detect attacks on existing recommender system.

**Model Question Paper**

**QP CODE:**

**Reg No:** \_\_\_\_\_

**Name:** \_\_\_\_\_

**PAGES : 3**

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

**SIXTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR**

**Course Code: AIT312**

**Course Name: Recommendation System**

**Max. Marks : 100**

**Duration: 3 Hours**

**PART A**

**Answer All Questions. Each Question Carries 3 Marks**

1. List any two purposes of recommender systems.
2. Indicate the main idea of collaborative recommendation approaches.
3. Describe the two types of outputs generated with pure collaborative approaches that takes matrix of given user–item ratings as the only input .
4. Define case amplification. How it can be computed?
5. Explain the need of item-based nearest neighbor recommendation system.
6. Define classical constraint satisfaction problem (CSP).
7. Differentiate between parallelized hybridization design and pipelined hybridization design with suitable diagram.
8. What is meant by monolithic hybridization design?
9. Differentiate between internal validity and external validity.
10. Specify the effect of base recommendation algorithm.

**(10x3=30)**

**Part B**

(Answer any one question from each module. Each question carries 14 Marks)

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11. (a) Describe about user-based nearest neighbor recommendation system which deals with new items for which no ratings exist. (7)
- (b) Explain about Rocchio's relevance feedback method (7)

OR

12. (a) Summarize the implicit and explicit rating mechanism in collaborative recommendation approaches. (7)
- (b) Explain any two techniques that deal with data sparsity and the cold-start problem. (7)
13. (a) Explain *QuickXPlain* algorithm that calculates one conflict set at a time for a given set of constraints. (7)
- (b) Which are the ways available to specify defaults? Explain how derived defaults can be determined. (7)

OR

14. (a) Explain about the ranking of items/utility-based recommendation. (7)
- (b) Describe *DynamicCritiquing* algorithm. (7)
15. (a) Explain about feature combination hybrids. (7)
- (b) Describe feature augmentation hybrid. (7)

OR

16. (a) Explain about different parallelized hybridization strategies. (7)
- (b) Describe pipelined hybridization design methods. (7)
17. (a) Explain about offline and online evaluations in recommender systems. (7)
- (b) Describe the general goals of evaluation design. (7)

OR

18. (a) Discuss about the design issues in offline recommender evaluation. Illustrate with a case study. (7)
- (b) Explain about accuracy metrics in offline evaluation. (7)
19. (a) How do you quantify attack impact on recommender system? (7)

- (b) Discuss about different attacks on recommender system. (7)

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OR

20. (a) Discuss about different methods available to detect attacks on recommender system. (7)

- (b) Explain how to design robust recommendation algorithms. (7)

| Teaching Plan                                       |                                                                                                                    |                               |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------|
| No                                                  | Contents                                                                                                           | No. of Lecture Hours (36 hrs) |
| <b>Module - 1 (Introduction to basic concepts )</b> |                                                                                                                    | <b>(7 hours)</b>              |
| 1.1                                                 | Introduction to basic concepts and Recent developments                                                             | 1 hour                        |
| 1.2                                                 | Collaborative recommendation : User-based nearest neighbor recommendation                                          | 1 hour                        |
| 1.3                                                 | Collaborative recommendation : Item-based nearest neighbor recommendation                                          | 1 hour                        |
| 1.4                                                 | Collaborative recommendation ratings                                                                               | 1 hour                        |
| 1.5                                                 | Model-based and preprocessing-based approaches, Recent practical approaches and systems                            | 1 hour                        |
| 1.6                                                 | Content-based recommendation - Content representation and content similarity                                       | 1 hour                        |
| 1.7                                                 | Similarity-based retrieval and Other text classification methods                                                   | 1 hour                        |
| <b>Module - 2 (Knowledge-based recommendation)</b>  |                                                                                                                    | <b>(8 hours)</b>              |
| 2.1                                                 | Knowledge representation and reasoning , Constraints, Cases and similarities                                       | 1 hour                        |
| 2.2                                                 | Interacting with constraint-based recommenders - Defaults                                                          | 1 hour                        |
| 2.3                                                 | Dealing with unsatisfiable requirements and empty result sets                                                      | 1 hour                        |
| 2.4                                                 | Proposing repairs for unsatisfiable requirements                                                                   | 1 hour                        |
| 2.5                                                 | Ranking the items/utility-based recommendation                                                                     | 1 hour                        |
| 2.6                                                 | Interacting with case-based recommenders Introduction                                                              | 1 hour                        |
| 2.7                                                 | Critiquing -Compound critiquing                                                                                    | 1 hour                        |
| 2.8                                                 | Dynamic critiquing                                                                                                 | 1 hour                        |
| <b>Module- 3 (Hybrid recommendation)</b>            |                                                                                                                    | <b>(7 hours)</b>              |
| 3.1                                                 | Hybrid recommendation approaches - Opportunities for hybridization Recommendation paradigms, Hybridization designs | 1 hour                        |
| 3.2                                                 | Monolithic hybridization design - Feature combination hybrids                                                      | 1 hour                        |
| 3.3                                                 | Feature augmentation hybrids                                                                                       | 1 hour                        |
| 3.4                                                 | Parallelized hybridization design -Mixed hybrids, Switching hybrids                                                | 1 hour                        |
| 3.5                                                 | Weighted hybrids                                                                                                   | 1 hour                        |



|                                                          |                                                                                                                                                                                                                                                                     |                  |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.6                                                      | Pipelined hybridization design Cascade hybrids, Meta-level hybrids                                                                                                                                                                                                  | 1 hour           |
| 3.7                                                      | Limitations of hybridization strategies                                                                                                                                                                                                                             | 1 hour           |
| <b>Module - 4 ( Evaluating Recommender Systems )</b>     |                                                                                                                                                                                                                                                                     | <b>(8 hours)</b> |
| 4.1                                                      | Introduction - Evaluation Paradigms , User Studies , Online Evaluation<br>Offline Evaluation with Historical Data Sets                                                                                                                                              | 1 hour           |
| 4.2                                                      | Goals of Evaluation Design - Accuracy, Coverage , Confidence and Trust ,<br>Novelty                                                                                                                                                                                 | 1 hour           |
| 4.3                                                      | General Goals of Evaluation Design - Serendipity ,Diversity , Robustness and<br>Stability Scalability                                                                                                                                                               | 1 hour           |
| 4.4                                                      | Design Issues in Offline Recommender Evaluation - Case Study of the<br>Netflix Prize Data Set                                                                                                                                                                       | 1 hour           |
| 4.5                                                      | Design Issues in Offline Recommender Evaluation -Segmenting the Ratings for<br>Training and Testing - Hold-Out , Cross-Validation , Comparison with<br>Classification                                                                                               | 1 hour           |
| 4.6                                                      | Accuracy Metrics in Offline Evaluation - Measuring the Accuracy of Ratings<br>Prediction , RMSE versus MAE, Impact of the Long Tail                                                                                                                                 | 1 hour           |
| 4.7                                                      | Evaluating Ranking via Correlation , Evaluating Ranking via Utility<br>Evaluating Ranking via Receiver Operating Characteristic                                                                                                                                     |                  |
| 4.8                                                      | Limitations of Evaluation Measures - Avoiding Evaluation Gaming                                                                                                                                                                                                     | 1 hour           |
| <b>Module- 5 (Attack-Resistant Recommender Systems )</b> |                                                                                                                                                                                                                                                                     | <b>(6 hours)</b> |
| 5.1                                                      | Introduction Understanding the Trade-Offs in Attack Models - Quantifying<br>Attack Impact                                                                                                                                                                           | 1 hour           |
| 5.2                                                      | Types of Attacks - Random Attack . Average Attack Bandwagon                                                                                                                                                                                                         | 1 hour           |
| 5.3                                                      | Reverse Bandwagon Attack , Probe Attack Segment Attack , Effect of Base<br>Recommendation Algorithm                                                                                                                                                                 | 1 hour           |
| 5.4                                                      | Detecting Attacks on Recommender Systems - Individual Attack Profile<br>Detection ,Group Attack Profile Detection - Preprocessing Methods Online<br>Methods                                                                                                         | 1 hour           |
| 5.5                                                      | Strategies for Robust Recommender Design - Preventing Automated Attacks<br>with CAPTCHAs Using Social Trust . Designing Robust Recommendation<br>Algorithms - Incorporating Clustering in Neighborhood Methods Fake<br>Profile Detection during Recommendation Time | 1 hour           |
| 5.6                                                      | Association-Based Algorithms                                                                                                                                                                                                                                        | 1 hour           |



| AIT322 | CONCEPTS IN<br>COMPUTER GRAPHICS<br>AND IMAGE<br>PROCESSING | Category | L | T | P | Credit | Year of<br>Introduction |
|--------|-------------------------------------------------------------|----------|---|---|---|--------|-------------------------|
|        |                                                             | PEC      | 2 | 1 | 0 | 4      | 2019                    |

**Preamble:** The purpose of this course is to make awareness about strong theoretical relationships between computer graphics and image processing. This course helps the learner to understand three-dimensional environment representation in a computer, transformation of 2D/3D objects, basic mathematical techniques and algorithms used to build useful applications, imaging, and image processing techniques. The study of computer graphics and image processing develops the ability to create image processing frameworks for different domains and develops algorithms for emerging display technologies.

**Prerequisite:** A sound knowledge of Mathematics and a programming language.

**Course Outcomes:** After the completion of the course the student will be able to

| CO# | CO                                                                                                                                                         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Describe the working principles of graphics devices(Cognitive Knowledge level: Understand)                                                                 |
| CO2 | Illustrate line drawing, circle drawing and polygon filling algorithms(Cognitive Knowledge level: Apply)                                                   |
| CO3 | Demonstrate geometric representations, transformations on 2D & 3D objects, clipping algorithms and projection algorithms(Cognitive Knowledge level: Apply) |
| CO4 | Summarize visible surface detection methods(Cognitive Knowledge level: Understand)                                                                         |
| CO5 | Summarize the concepts of digital image representation, processing and demonstrate pixel relationships(Cognitive Knowledge level: Apply)                   |
| CO6 | Solve image enhancement and segmentation problems using spatial domain techniques(Cognitive Knowledge level: Apply)                                        |